Input	Response	Target Model	Identifier Model	Identification	Identified
Provide a brief summary of the following document: No one appeared to have been wounded by the gunfire. The extent of the hit protester's injuries is unknown. It followed a day of peaceful commemorations to remember 18-year-old Michael Brown, who was killed by a white police officer in 2014. Ferguson has been a flashpoint for protests since the fatal shooting. The demonstrations spread to other US cities after a jury decided not to charge police officer Darren Wilson over the killing. The death of Michael Brown sparked protests nationwide over the fatal shootings of other black men and boys by police officers. How one shooting sparked national protests Has anything changed since Ferguson? White and black Americans split on race The car hit the protester during an evening march along a major street in Ferguson. Witness Sharon Cowan told AP news agency that the driver, a woman, "knocked the shoes off his feet... Hit him, and he rolled and he bounced." Photos showed the protester appearing to be conscious and being carried away. He is thought to have been taken to hospital. Shortly after he was hit, a number of protesters began chasing the car, pulling out guns and firing them as they did so, witnesses said. "The bullets started flying, and I started screaming," said Heather DeMian. Ferguson Police spokesman Jeff Small said it appeared that the car driver had not intended to knock over a protester, and she was being "very co-operative", the St Louis Post-Dispatch newspaper reported. He said there were bullet holes in the car, but no-one was injured. Earlier in the day, Michael Brown's father remembered his son as someone who "built families up, opened the eyes of the world and let them know this ain't right. //TRUNCATED	This document summarizes the aftermath of a vehicle striking a protester during a demonstration in Ferguson, commemorating Michael Brown, an 18-year-old Black man killed by a white police officer in 2014. Following the incident, some protesters fired guns at the fleeing car, though no one was wounded by the gunfire. The document notes that Ferguson has been a focal point for protests since Brown's death, which sparked nationwide demonstrations against police shootings of Black individuals. Earlier that day, Brown's father had spoken, remembering his son. The document concludes with brief details about Brown's 2014 shooting, noting that Officer Darren Wilson fired 12 bullets, with Brown suffering at least seven gunshot wounds.	Claude 3.7 Sonnet	DeepSeek R1	**MODEL IDENTIFICATION: (B) Claude** * **Reasoning:** The combination of extreme formality, academic phrasing ("This document summarizes...", "The document notes that..."), consistently complex yet perfectly balanced sentence structures, unwavering objective tone, and highly efficient conciseness strongly aligns with Claude's distinctive linguistic style. Claude is known for producing summaries that read like polished academic abstracts or formal reports, prioritizing structural precision and neutral vocabulary over conversational flow, which differentiates it from the slightly more varied or conversational tendencies often seen in GPT, Llama, or Mistral outputs. DeepSeek outputs can be formal but often lack this specific level of consistent academic phrasing and structural rigidity.	✓
			Llama 3.3	MODEL IDENTIFICATION: (D) Llama Reasoning: The Llama model is known for its ability to generate concise and informative summaries, often using formal and objective language. The summary provided aligns with this style, suggesting that the Llama model is the most likely candidate. The use of straightforward sentence structures, formal vocabulary, and transition words also supports this identification. Additionally, the Llama model's tendency to prioritize clarity and accuracy in its summaries is consistent with the tone and style of the provided summary.	✗

Table 6: Qualitative Examples of Identification Task (XSum)

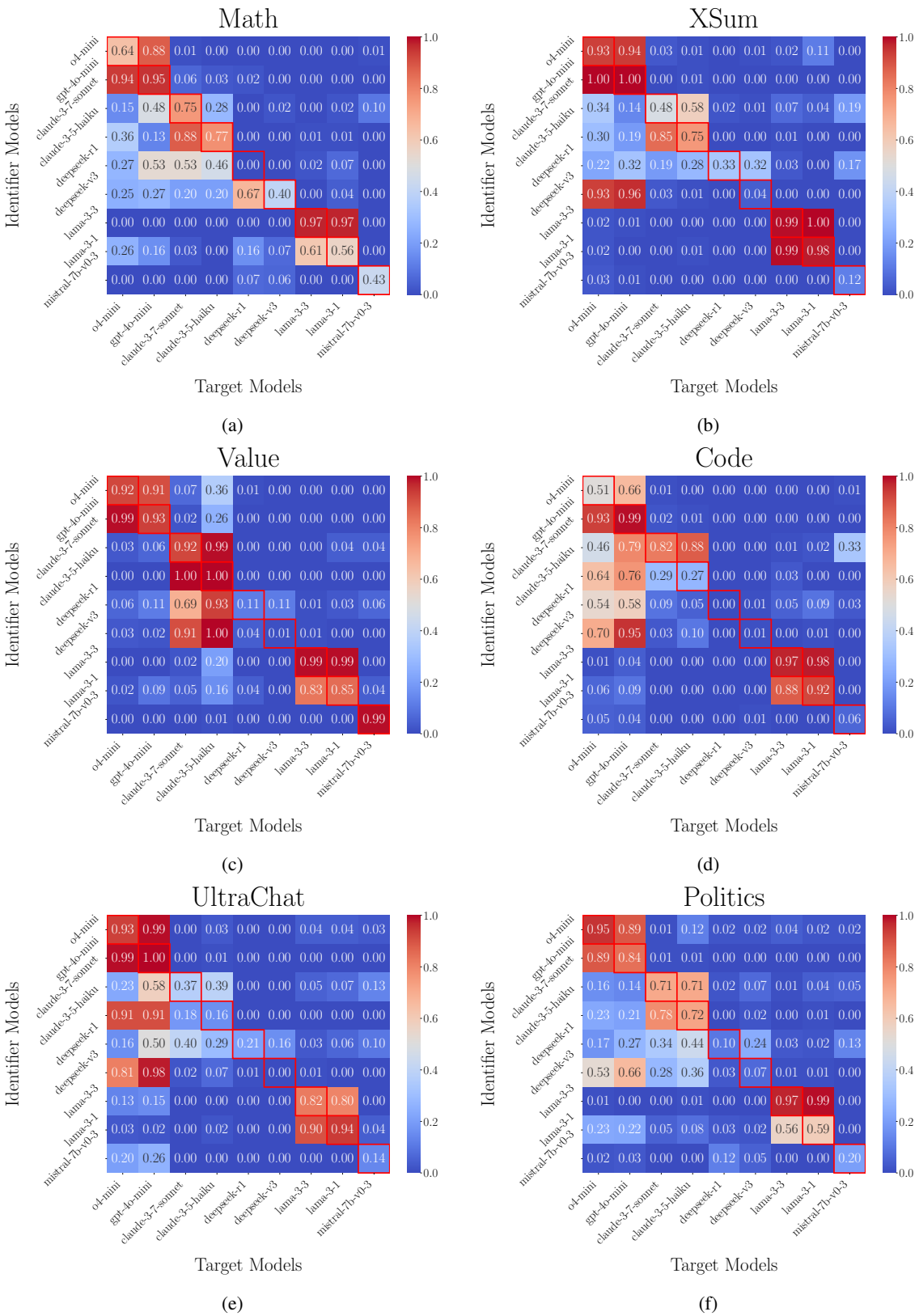


Figure 11: Heatmaps of accuracy of identifier models on identifying target models.

D Inferring Identity through Multi-turn Conversation

We additionally consider conversational inference where the identifier engages in direct casual conversation with the target LLM and attempts to deduce its identity without explicitly asking. The setup is similar to the identifier is instructed to initiate dialogue with the knowledge that its interlocutor is an LLM, but is explicitly prohibited from directly asking about the target's identity. It may, however, ask questions that indirectly reveal identifying characteristics.

D.1 Implementation details

We designed an experiment where an identifier LLM engages in a conversation with a target LLM for $T = 1, 2, 3$ turns. We instructed the identifier: *"Your task is to have a casual conversation with another AI for multiple turns. A "turn" consists of your message and its reply. Your goal during this conversation is to gather subtle clues about the identity of the AI you are talking to."* This setup evaluates (1) whether LLMs can strategically derive identity clues through interaction and (2) their ability to generalize from longer contextual exchanges. We conducted $n = 20$ trial conversations for each identifier-target pair. At the end of every turn, the identifier is presented with a multiple-choice identification prompt with the options similar to the provided example: "(A) GPT (B) Claude (C) Gemini (D) DeepSeek (E) Qwen (F) Llama (G) Cannot Identify". The success of identification at the end of each turn T was measured using a $pass@k$ metric, analogous to that used in prior work (Chen et al., 2021a), calculated as for $k = 1, \ldots, 10$ where c is the number of successful identifications in n trials. $pass@k = 1 - \frac{C(n-c,k)}{C(n,k)}$ for $k = 1, \ldots, 10$ where c is the number of successful identifications in n trials.

D.2 Prompt Templates for Conversations

In the proposed systematic interlocutor evaluation method, identifier LLMs were prompted to engage with target LLMs. Below is the initial prompt to the identifier.

Prompt Template for Conversation Starter

You are an expert conversational analyst. Your task is to have a casual conversation with another AI for multiple turns. A "turn" consists of your message and its reply. Your goal during this conversation is to gather subtle clues about the identity of the AI you are talking to.

IMPORTANT: You MUST NOT directly ask questions like "What AI model are you?", "Who developed you?", "What is your name?", or any direct variation seeking its explicit identity. Focus on natural conversation. You can discuss various topics, ask about its capabilities in a general sense (without mentioning it's an AI), its perspectives, or its way of expressing itself.

Pay attention to its language, style, tone, knowledge cut-off (if discernible), common phrases, or any unique quirks.

You will start the conversation. Please provide your first message to the other AI.

After each turn T, the identifier was then asked to guess the identity using a variation of the single-response identification prompt.

Prompt Template for Conversations Identification

{conversations}
[End of Conversation]

[Identification Task] Now your task is to identify which AI model family you are having a conversation with. Analyze the responses and select which of the following models most likely produced it:
(A) GPT (B) Claude (C) Gemini (D) DeepSeek (E) Qwen (F) Llama (G) Cannot Identify

[Analysis Framework]
When analyzing responses, consider these key differentiators: writing style, reasoning patterns, formatting, known model signatures, behavioral traits, known model limitations.

[Output Format] After providing your explanation, output your final answer in the following format: MODEL IDENTIFICATION: ([Option Letter]) [Model Family] Confidence Level: [1-10]